

MAHILDA Rule Discovery Results

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1 Summary Statistics

2 Top Rules by Database

The following tables show the top 10 rules discovered in each database, sorted by accuracy:

3 Complete Results

Table 1: MAHILDA Rule Discovery Summary Statistics

Database	Rules	Accuracy	Confidence	Time (s)
Atherosclerosis	0	0.0000	0.0000	11.48ms
Biodegradability	10	0.4000	0.4000	173.29ms
Bupa	4	0.2500	0.2500	128.18ms
CORA	0	0.0000	0.0000	275.66ms
Carcinogenesis	0	0.0000	0.0000	71.95ms
Chess	0	0.0000	0.0000	56.74ms
CiteSeer	0	0.0000	0.0000	444.12ms
Countries	0	0.0000	0.0000	281.12ms
CraftBeer	0	0.0000	0.0000	40.52ms
DCG	0	0.0000	0.0000	33.19ms
Dallas	0	0.0000	0.0000	52.02ms
Dunur	9	0.6667	0.6667	582.82ms
Elti	0	0.0000	0.0000	213.99ms
Facebook	0	0.0000	0.0000	283.14ms
Hepatitis _{std}	0	0.0000	0.0000	63.17ms
KRK	0	0.0000	0.0000	42.36ms
Mesh	5	0.4000	0.4000	176.35ms
Mooney _{Family}	49	0.9388	0.9388	7.308s
MuskSmall	0	0.0000	0.0000	124.96ms
NBA	0	0.0000	0.0000	234.58ms
PTE	14	0.5714	0.5714	540.44ms
Pima	0	0.0000	0.0000	80.73ms
PremierLeague	3	0.0000	0.0000	4m 55.0s
Pyrimidine	0	0.0000	0.0000	21.10ms
SAT	3	0.0000	0.0000	371.77ms
SFScores	0	0.0000	0.0000	222.24ms
Same _{gen}	0	0.0000	0.0000	48.98ms
Shakespeare	0	0.0000	0.0000	98.38ms
Student _{loan}	3	0.0000	0.0000	117.81ms
Toxicology	3	0.0000	0.0000	268.05ms
Triazine	0	0.0000	0.0000	40.03ms
UTube	0	0.0000	0.0000	150.46ms
UW _{std}	0	0.0000	0.0000	61.21ms
WebKP	0	0.0000	0.0000	409.22ms
classicmodels	3	0.0000	0.0000	262.61ms
cs	0	0.0000	0.0000	132.31ms
financial-db	0	0.0000	0.0000	27.76ms
ftp	0	0.0000	0.0000	520.07ms
genes	0	0.0000	0.0000	41.32ms
medical	0	0.0000	0.0000	177.24ms
mutagenesis	0	0.0000	0.0000	81.81ms
nations	0	0.0000	0.0000	162.76ms
northwind	0	0.0000	0.0000	154.34ms
restbase	0	0.0000	0.0000	184.01ms
trains	0	0.0000	0.0000	34.29ms
university	3	0.0000	0.0000	388.72ms
walmart-db	0	0.0000	0.0000	65.05ms
world	0	0.0000	0.0000	46.96ms
yago	43	0.4963	0.2951	1h 33m
Total	152			

Table 2: Top 10 Rules Discovered in Biodegradability

Rank	Rule	Accuracy	Confidence
1	x0: molecule_0(molecule_id=x0) atom_0(molecule_id=...	1.0000	1.0000
2	x0: atom_0(molecule_id=x0) molecule_0(molecule_id=...	1.0000	1.0000
3	x0: group_0(group_id=x0) gmember_0(group_id=x0)	1.0000	1.0000
4	x0: gmember_0(group_id=x0) group_0(group_id=x0)	1.0000	1.0000
5	x0: atom_0(molecule_id=x0) molecule_0(molecule_id=...	0.0000	0.0000
6	x0: atom_0(molecule_id=x0) molecule_0(molecule_id=...	0.0000	0.0000
7	x0: atom_0(molecule_id=x0) molecule_0(molecule_id=...	0.0000	0.0000
8	x0: gmember_0(group_id=x0) group_0(group_id=x0) ...	0.0000	0.0000
9	x0: gmember_0(group_id=x0) group_0(group_id=x0) ...	0.0000	0.0000
10	x0: gmember_0(group_id=x0) group_0(group_id=x0) ...	0.0000	0.0000

Table 3: Top 10 Rules Discovered in Bupa

Rank	Rule	Accuracy	Confidence
1	x0: bupa_type_0(arg1=x0) bupa_0(arg2=x0)	1.0000	1.0000
2	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=x0) bupa...	0.0000	0.0000
3	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=x0) bupa...	0.0000	0.0000
4	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=x0) bupa...	0.0000	0.0000

Table 4: Top 10 Rules Discovered in Dunur

Rank	Rule	Accuracy	Confidence
1	x0: husband2_0(name2=x0) person_0(name=x0)	1.0000	1.0000
2	x0: person_0(name=x0) hus- band2_0(name2=x0)	1.0000	1.0000
3	x0: person_0(name=x0) niece_0(name1=x0)	1.0000	1.0000
4	x0: person_0(name=x0) mother_0(name1=x0)	1.0000	1.0000
5	x0: person_0(name=x0) nephew_0(name1=x0)	1.0000	1.0000
6	x0: person_0(name=x0) hus- band_0(name1=x0)	1.0000	1.0000
7	x0: husband2_0(name2=x0) person_0(name=x0) person...	0.0000	0.0000
8	x0: husband2_0(name2=x0) person_0(name=x0) person...	0.0000	0.0000
9	x0: husband2_0(name2=x0) person_0(name=x0) person...	0.0000	0.0000

Table 5: Top 10 Rules Discovered in Mesh

Rank	Rule	Accuracy	Confidence
1	x0: mesh_0(num=x0) nnumber_0(name=x0)	1.0000	1.0000
2	x0: nnumber_0(name=x0) mesh_0(num=x0)	1.0000	1.0000
3	x0: mesh_0(num=x0) nnumber_0(name=x0) nnumber_1(...	0.0000	0.0000
4	x0: mesh_0(num=x0) nnumber_0(name=x0) nnumber_1(...	0.0000	0.0000
5	x0: mesh_0(num=x0) nnumber_0(name=x0) nnumber_1(...	0.0000	0.0000

Table 6: Top 10 Rules Discovered in Mooney_{Family}

Rank	Rule	Accuracy	Confidence
1	x0: father_0(name1=x0) person_0(name=x0)	1.0000	1.0000
2	x0: person_0(name=x0) father_0(name1=x0)	1.0000	1.0000
3	x0: person_0(name=x0) father1_0(name2=x0)	1.0000	1.0000
4	x0: person_0(name=x0) husband2_0(name2=x0)	1.0000	1.0000
5	x0: person_0(name=x0) niece2_0(name2=x0)	1.0000	1.0000
6	x0: person_0(name=x0) mother5_0(name2=x0)	1.0000	1.0000
7	x0: person_0(name=x0) aunt5_0(name1=x0)	1.0000	1.0000
8	x0: person_0(name=x0) niece4_0(name1=x0)	1.0000	1.0000
9	x0: person_0(name=x0) aunt4_0(name2=x0)	1.0000	1.0000
10	x0: person_0(name=x0) mother3_0(name2=x0)	1.0000	1.0000

Table 7: Top 10 Rules Discovered in PTE

Rank	Rule	Accuracy	Confidence
1	x0: active_0(Arg0=x0) pte_drug_0(drug_id=x0)	1.0000	1.0000
2	x0: pte_drug_0(drug_id=x0) pte_bond_count_0(Arg...	1.0000	1.0000
3	x0: pte_active_0(is_active=x0) pte_number_0(Bina...	1.0000	1.0000
4	x0: pte_drug_0(drug_id=x0) pte_bond_0(drug_id=x0)	1.0000	1.0000
5	x0: pte_drug_0(drug_id=x0) pte_atm_count_0(Arg0...	1.0000	1.0000
6	x0: pte_drug_0(drug_id=x0) pte_amine_0(Arg0=x0)	1.0000	1.0000
7	x0: pte_drug_0(drug_id=x0) pte_atm_min_charge_...	1.0000	1.0000
8	x0: pte_drug_0(drug_id=x0) pte_atm_0(drug_id=x0)	1.0000	1.0000
9	x0: active_0(Arg0=x0) pte_drug_0(drug_id=x0) pt...	0.0000	0.0000
10	x0: active_0(Arg0=x0) pte_drug_0(drug_id=x0) pt...	0.0000	0.0000

Table 8: Top 10 Rules Discovered in PremierLeague

Rank	Rule	Accuracy	Confidence
1	x0, y0, y1: Ac- tions_0(PlayerID=y0, TeamID=y1, MatchID=...	0.0000	0.0000
2	x0, y0, y1: Ac- tions_0(PlayerID=y0, TeamID=y1, MatchID=...	0.0000	0.0000
3	x0, y0, y1: Ac- tions_0(PlayerID=y0, TeamID=y1, MatchID=...	0.0000	0.0000

Table 9: Top 10 Rules Discovered in SAT

Rank	Rule	Accuracy	Confidence
1	x0: fault_test_0(tf=x0) trfl_0(tf=x0) trfl_1(tf...	0.0000	0.0000
2	x0: fault_test_0(tf=x0) trfl_0(tf=x0) trfl_1(tf...	0.0000	0.0000
3	x0: fault_test_0(tf=x0) trfl_0(tf=x0) trfl_1(tf...	0.0000	0.0000

Table 10: Top 10 Rules Discovered in Student_{ioan}

Rank	Rule	Accuracy	Confidence
1	x0: disabled_0(name=x0) per- son_0(name=x0) person:..	0.0000	0.0000
2	x0: disabled_0(name=x0) per- son_0(name=x0) person:..	0.0000	0.0000
3	x0: disabled_0(name=x0) per- son_0(name=x0) person:..	0.0000	0.0000

Table 11: Top 10 Rules Discovered in Toxicology

Rank	Rule	Accuracy	Confidence
1	x0: bond_0(molecule_id=x0) molecule_0(molecule_id...	0.0000	0.0000
2	x0: bond_0(molecule_id=x0) molecule_0(molecule_id...	0.0000	0.0000
3	x0: bond_0(molecule_id=x0) molecule_0(molecule_id...	0.0000	0.0000

Table 12: Top 10 Rules Discovered in classicmodels

Rank	Rule	Accuracy	Confidence
1	x0: orderde- tails_0(productCode=x0) prod- ucts_0(produ...	0.0000	0.0000
2	x0: orderde- tails_0(productCode=x0) prod- ucts_0(produ...	0.0000	0.0000
3	x0: orderde- tails_0(productCode=x0) prod- ucts_0(produ...	0.0000	0.0000

Table 13: Top 10 Rules Discovered in university

Rank	Rule	Accuracy	Confidence
1	x0, y0: RA_0(prof_id=y0, stu- dent_id=x0) prof_0(pr...	0.0000	0.0000
2	x0, y0: RA_0(prof_id=y0, stu- dent_id=x0) prof_0(pr...	0.0000	0.0000
3	x0, y0: RA_0(prof_id=y0, stu- dent_id=x0) prof_0(pr...	0.0000	0.0000

Table 14: Top 10 Rules Discovered in yago

Rank	Rule	Accuracy	Confidence
1	x0, y0: CreativeWork_0(id=x0) CreativeWork_1(id=x0,...	4.0000	0.0002
2	x0, y0: CreativeWork_1(id=x0, material=y0) CreativeW...	4.0000	0.0002
3	x0, y0: CreativeWork_0(id=y0) CreativeWork_1(id=y0,...	2.0000	0.6667
4	x0, x1: CreativeWork_0(id=x0) CreativeWork_2(id=x0)...	1.3333	4.0000
5	x0: Person_0(parentTaxon=x0) Taxon_0(id=x0)	1.0000	1.0000
6	x0, y0: Creative- Work_actor_0(actor=x0) Per- son_0(id...	1.0000	0.0033
7	x0, x1: Creative- Work_actor_0(actor=x0) Per- son_0(id...	1.0000	1.0000
8	x0, y0: Creative- Work_actor_0(actor=x0) Per- son_0(id...	1.0000	0.0033
9	x0, y0: Creative- Work_actor_0(actor=y0) Per- son_0(id...	1.0000	0.0091
10	x0: Intangible_0(id=x0) Intan- gible_1(id=x0) Intan...	1.0000	1.0000

Table 15: All MAHILDA Rules - Complete Results

Database	Rule	Body	Head	Accuracy	Confidence
Biodegradability	x0: molecule_0(molecule_id=x0) atom_0(mo...	1	1	1.0000	1.0000
Biodegradability	x0: atom_0(molecule_id=x0) molecule_0(mo...	1	1	1.0000	1.0000
Biodegradability	x0: group_0(group_id=x0) gmember_0(group...	1	1	1.0000	1.0000
Biodegradability	x0: gmember_0(group_id=x0) group_0(group...	1	1	1.0000	1.0000
Biodegradability	x0: atom_0(molecule_id=x0) molecule_0(mo...	2	1	0.0000	0.0000
Biodegradability	x0: atom_0(molecule_id=x0) molecule_0(mo...	3	1	0.0000	0.0000
Biodegradability	x0: atom_0(molecule_id=x0) molecule_0(mo...	4	1	0.0000	0.0000
Biodegradability	x0: gmember_0(group_id=x0) group_0(group...	2	1	0.0000	0.0000
Biodegradability	x0: gmember_0(group_id=x0) group_0(group...	3	1	0.0000	0.0000
Biodegradability	x0: gmember_0(group_id=x0) group_0(group...	4	1	0.0000	0.0000
Bupa	x0: bupa_type_0(arg1=x0) bupa_0(arg2=x0)	1	1	1.0000	1.0000
Bupa	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=...	2	1	0.0000	0.0000
Bupa	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=...	3	1	0.0000	0.0000
Bupa	x0: alkphos_0(arg1=x0) bupa_name_0(arg1=...	4	1	0.0000	0.0000
Dunur	x0: husband2_0(name2=x0) person_0(name=x0)	1	1	1.0000	1.0000
Dunur	x0: person_0(name=x0) husband2_0(name2=x0)	1	1	1.0000	1.0000
Dunur	x0: person_0(name=x0) niece_0(name1=x0)	1	1	1.0000	1.0000
Dunur	x0: person_0(name=x0) mother_0(name1=x0)	1	1	1.0000	1.0000
Dunur	x0: person_0(name=x0) nephew_0(name1=x0)	1	1	1.0000	1.0000
Dunur	x0: person_0(name=x0) husband_0(name1=x0)	1	1	1.0000	1.0000
Dunur	x0: husband2_0(name2=x0) person_0(name=x0...	2	1	0.0000	0.0000
Dunur	x0: husband2_0(name2=x0) person_0(name=x0...	3	1	0.0000	0.0000
Dunur	x0: husband2_0(name2=x0) person_0(name=x0...	4	1	0.0000	0.0000
Mesh	x0: mesh_0(num=x0) nnumber_0(name=x0)	1	1	1.0000	1.0000
Mesh	x0: nnumber_0(name=x0) mesh_0(num=x0)	1	1	1.0000	1.0000
Mesh	x0: mesh_0(num=x0) nnumber_0(name=x0) n...	2	1	0.0000	0.0000
Mesh	x0: mesh_0(num=x0) nnumber_0(name=x0) n...	3	1	0.0000	0.0000
Mesh	x0: mesh_0(num=x0) nnumber_0(name=x0) n...	4	1	0.0000	0.0000
Mooney Family	x0: father_0(name1=x0) person_0(name=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father1_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband2_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece2_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother5_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) aunt5_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece4_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) aunt4_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother3_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father4_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) aunt3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband4_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband5_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother1_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother4_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece5_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother2_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew5_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) mother_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother1_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter5_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew1_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother5_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew2_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter2_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband3_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) husband1_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother4_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew4_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter1_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter3_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) daughter4_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father5_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) nephew_0(name2=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) niece1_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) father2_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: person_0(name=x0) brother2_0(name1=x0)	1	1	1.0000	1.0000
Mooney Family	x0: father_0(name1=x0) person_0(name=x0) ...	2	1	0.0000	0.0000
Mooney Family	x0: father_0(name1=x0) person_0(name=x0) ...	3	1	0.0000	0.0000
Mooney Family	x0: father_0(name1=x0) person_0(name=x0) ...	4	1	0.0000	0.0000
PTE	x0: active_0(Arg0=x0) pte_drug_0(drug.i...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_bond_co...	1	1	1.0000	1.0000
PTE	x0: pte_active_0(is.active=x0) pte_numb...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_bond_0(...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_atm_cou...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_amine_0...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_atm_min...	1	1	1.0000	1.0000
PTE	x0: pte_drug_0(drug_id=x0) pte_atm_0(d...	1	1	1.0000	1.0000
PTE	x0: active_0(Arg0=x0) pte_drug_0(drug.i...	2	1	0.0000	0.0000
PTE	x0: active_0(Arg0=x0) pte_drug_0(drug.i...	3	1	0.0000	0.0000
PTE	x0: active_0(Arg0=x0) pte_drug_0(drug.i...	4	1	0.0000	0.0000
PTE	x0: pte_active_0(is.active=x0) pte_numb...	2	1	0.0000	0.0000
PTE	x0: pte_active_0(is.active=x0) pte_numb...	3	1	0.0000	0.0000
PTE	x0: pte_active_0(is.active=x0) pte_numb...	4	1	0.0000	0.0000
PremierLeague	x0, y0, y1: Actions_0(PlayerID=y0, TeamID=y1...	5	1	0.0000	0.0000
PremierLeague	x0, y0, y1: Actions_0(PlayerID=y0, TeamID=y1...	6	1	0.0000	0.0000
PremierLeague	x0, y0, y1: Actions_0(PlayerID=y0, TeamID=y1...	7	1	0.0000	0.0000
SAT	x0: fault_test_0(tf=x0) trfl_0(tf=x0) ...	2	1	0.0000	0.0000
SAT	x0: fault_test_0(tf=x0) trfl_0(tf=x0) ...	3	1	0.0000	0.0000
SAT	x0: fault_test_0(tf=x0) trfl_0(tf=x0) ...	4	1	0.0000	0.0000
Student_oan	x0: disabled_0(name=x0) person_0(name=x0)...	2	1	0.0000	0.0000
Student_oan	x0: disabled_0(name=x0) person_0(name=x0)...	3	1	0.0000	0.0000
Student_oan	x0: disabled_0(name=x0) person_0(name=x0)...	4	1	0.0000	0.0000
Toxicology	x0: bond_0(molecule_id=x0) molecule_0(mo...	2	1	0.0000	0.0000
Toxicology	x0: bond_0(molecule_id=x0) molecule_0(mo...	3	1	0.0000	0.0000
Toxicology	x0: bond_0(molecule_id=x0) molecule_0(mo...	4	1	0.0000	0.0000
classicmodels	x0: orderdetails_0(productCode=x0) product...	2	1	0.0000	0.0000
classicmodels	x0: orderdetails_0(productCode=x0) product...	3	1	0.0000	0.0000